Department of Computer Science and Engineering

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(VI Semester)

A Project Report on

DETECTION OF DIABETES USING MACHINE LEARNING

Submitted for the partial fulfilment of Project Centric Learning activity of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project work titled **" DETECTION OF DIABETES USING MACHINE LEARNING"** is carried out by Yedoti Premchand (23BTRCT074), Sai Sanjay (23BTRCT011), Tejaswanth (23BTRCT133), SalvaAbilash (23BTRCT117), Rithika Rv (23BTRCT112), Yaswanth Kumar (23BTRCT111) **,** a Bonafide students of Bachelor of Technology at the School of Computer Science and Engineering - CSE , Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfillment for the award of degree in Bachelor Technology in Computer Science & Engineering - CSE, during the year **2025-2026**.

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DECLARATION

We,**1.Yedoti.Premchand(23BTRCT074),2.Saisanjay(23BTRCT011), 3.Tejaswanth(23BTRCT133),4.SalvaAbilash(23BTRCT117), 5.Rithika Rv (23BTRCT112),6. Yaswanth Kumar (23BTRCT111)** are student's of 4th semester B.Tech in **Computer Science & Engineering**, JAIN(Deemed-to-be University), hereby declare that the project centric learning work titled **"DETECTION OF DIABETES USING MACHINE LEARNING"** has been carried out by us and submitted in partial fulfilment for the project centric learning activity of degree in **Bachelor of Technology in Computer Science & Engineering** during the academic year **2023-2027**.

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Abstract
Diabetes is one of the fastest-growing chronic diseases worldwide, affecting millions of people and increasing the risk of severe health complications such as heart disease, kidney failure, nerve damage, and vision problems. Early diagnosis plays a vital role in reducing these complications and improving patient health outcomes. Traditional diabetes detection methods generally require blood sample testing, repeated hospital visits, and continuous monitoring, which can be difficult for people living in rural or underdeveloped areas.
This research presents a machine learning-based diabetes detection system using Kernel Principal Component Analysis (Kernel PCA) and multiple classification algorithms. A dataset containing 521 patient records with 17 health-related attributes was used in this study. Data preprocessing techniques such as missing value handling, normalization, and dimensionality reduction were applied before training the models.
A total of 15 machine learning classification algorithms were tested under different train-test split ratios and threshold conditions. The models were evaluated using performance metrics such as accuracy, precision, recall, F1-score, and true negative rate. Among all models, the Kernel Support Vector Machine (KSVM) with a linear kernel achieved the highest accuracy of 100% using an 80-20 train-test split. Other high-performing models included XGBoost and CatBoost, both achieving 99.04% accuracy.
The results demonstrate that machine learning techniques can significantly improve early diabetes detection and assist healthcare professionals in making faster and more accurate decisions. Future work may involve integrating the system into hospital management systems and electronic health records for real-time medical applications.

Keywords: Diabetes Detection, Machine Learning, Kernel PCA, KSVM, CatBoost, XGBoost, Classification Algorithms, Healthcare Analytics, Early Diagnosis.

Chapter 1: Introduction
1.1 Background
Diabetes is a chronic disease that occurs when the body cannot properly regulate blood sugar levels. According to global health reports, diabetes cases are increasing rapidly every year. If diabetes is not identified early, it can lead to severe complications such as cardiovascular disease, kidney damage, blindness, and nerve-related disorders.
Traditional diagnostic methods involve laboratory testing and regular hospital visits. These procedures may be expensive, time-consuming, and difficult to access for people living in remote regions. Therefore, there is a growing need for intelligent systems that can support healthcare professionals in identifying diabetes quickly and accurately.
Machine Learning (ML) has become an important technology in the healthcare field. ML algorithms can analyze medical data, identify hidden patterns, and predict diseases with high accuracy. By using patient health information such as age, BMI, glucose level, and blood pressure, machine learning models can help doctors make faster decisions.

1.2 Problem Statement
Many diabetes detection systems still depend on traditional testing methods that require blood samples and repeated medical examinations. These methods are not always practical in areas with limited healthcare infrastructure.
The major problems addressed in this research are:
- Difficulty in early diabetes diagnosis
- High dependency on laboratory testing
- Limited accessibility to healthcare services
- Need for more accurate and automated prediction systems

1.3 Objectives
The main objectives of this study are:
- To develop a machine learning-based diabetes detection system.
- To compare the performance of 15 different classification algorithms.
- To improve prediction accuracy using Kernel PCA.
- To identify the best-performing algorithm for diabetes detection.
- To assist healthcare professionals in early diagnosis.

1.4 Scope of the Study
This study focuses on applying machine learning algorithms to predict diabetes using patient health data. The research mainly evaluates classification accuracy under different train-test split conditions and compares the effectiveness of various models.

1.5 Major Contributions
The major contributions of this work include:
- Implementation of 15 machine learning classification models.
- Application of Kernel PCA for dimensionality reduction.
- Comparison of model performance using different train-test splits.
- Achievement of 100% accuracy using KSVM with a linear kernel.
- Development of an efficient framework for diabetes prediction.

Chapter 2: Literature Survey

Machine learning techniques are widely used in healthcare applications for disease prediction and medical analysis. Several researchers have applied algorithms such as Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbor (KNN), Decision Tree (DT), Artificial Neural Networks (ANN), and boosting techniques for diabetes prediction.

Pragya Shrivastava et al. conducted a comprehensive review on diabetes prediction using machine learning techniques. Their study analyzed multiple algorithms including RF, SVM, ANN, and DT. The reported accuracy ranged between 80% and 88%.

Danish Ather et al. applied Support Vector Machine for diabetes prediction and achieved an accuracy of 86%.

Mansvi Daigavhane et al. compared different machine learning algorithms such as SVM, Random Forest, Naive Bayes, KNN, and Decision Tree. Their results showed that Random Forest achieved the best accuracy of 89%.

Nicole D’Souza et al. implemented Decision Tree, Random Forest, Naive Bayes, and ANN models for diabetes reduction. Their Decision Tree model achieved an accuracy of 98.2%.

Bukola Badeji-Ajisafe et al. used supervised learning methods such as Logistic Regression, SVM, KNN, and Random Forest for early diabetes detection. Among these, Random Forest achieved the highest performance with 87% accuracy.

Aditya C. Kulkarni and B.V.S. Satyasrikar studied the effect of preprocessing techniques on diabetes prediction using algorithms such as SVM, KNN, Naive Bayes, Decision Tree, and Random Forest. Their SVM model with PCA achieved 79.22% accuracy.

The literature survey indicates that machine learning methods significantly improve diabetes prediction accuracy. However, selecting suitable preprocessing methods and classification algorithms remains a major challenge.

Chapter 3: Proposed Methodology

3.1 Dataset Description

The dataset used in this research contains 521 patient records with 17 health-related attributes. Each patient record is labeled as:

1 = Diabetic

0 = Non-Diabetic

The dataset includes features such as:

Age

Gender

BMI

Blood pressure

Glucose level

Medical measurements

3.2 Data Preprocessing

Before training the machine learning models, the dataset was preprocessed using the following steps:

3.2.1 Handling Missing Values

Missing values in the dataset were replaced using average values to ensure data consistency.

3.2.2 Data Normalization

The dataset was normalized to improve model performance and reduce feature imbalance.

3.2.3 Dimensionality Reduction using Kernel PCA

Kernel Principal Component Analysis (Kernel PCA) was used to reduce dataset complexity and improve computational efficiency. The original dataset features were transformed into 7 important principal components.

3.3 Train-Test Splitting

The dataset was divided into different train-test ratios:

80% Training and 20% Testing

60% Training and 40% Testing

40% Training and 60% Testing

Balanced sampling was maintained to ensure equal representation of diabetic and non-diabetic patients.

3.4 Machine Learning Algorithms Used

A total of 15 machine learning algorithms were tested:

AdaBoost, CatBoost, Decision Tree Classifier (DTC), Gradient Boosting Classifier (GBC), K-Nearest Neighbor (KNN), Kernel Support Vector Machine (KSVM), Linear Discriminant Analysis (LDA), Logistic Regression (LG), LightGBM (LGBM), Linear SVM (LSVM), MLP Classifier (MLPC), Naive Bayes (NB), Quadratic Discriminant Analysis (QDA), Random Forest Classifier (RFC), and XGBoost (XGB)

3.5 Performance Metrics

The models were evaluated using the following performance metrics:

Accuracy:

Accuracy measures the percentage of correctly classified predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:
Precision measures how many positive predictions are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

Recall:
Recall measures the ability of the model to correctly identify positive cases.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:
F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Chapter 4: Results and Discussion

4.1 Overall Experimental Analysis

All 15 machine learning models were tested under the same experimental conditions to ensure fair comparison. Each model was trained and evaluated using polynomial and linear kernel settings with multiple train-test split configurations.

The overall results indicate that the 80-20 train-test split consistently provided the highest accuracy for most models.

4.2 CatBoost Results

The CatBoost classifier achieved a maximum accuracy of 99.04% using the polynomial kernel with an 80-20 train-test split. The model demonstrated strong stability across different testing conditions.

4.3 Decision Tree Classifier Results

The Decision Tree Classifier achieved its highest accuracy of 95.19% under the 80-20 split configuration for both polynomial and linear kernels.

4.4 Gradient Boosting Classifier Results

The Gradient Boosting Classifier reached a peak accuracy of 96.15% with both kernel types at the 80-20 split.

4.5 K-Nearest Neighbor Results

The KNN classifier achieved 98.08% accuracy using the linear kernel with an 80-20 split.

4.6 Kernel Support Vector Machine Results

The Kernel Support Vector Machine (KSVM) achieved the best overall performance with 100% accuracy using the linear kernel under the 80-20 train-test split.

The polynomial kernel also performed strongly with 99.04% accuracy.

4.7 Linear Discriminant Analysis Results

The LDA classifier achieved a maximum accuracy of 93.27% using the linear kernel.

4.8 Logistic Regression Results

Logistic Regression achieved 95.19% accuracy with both kernel types under the 80-20 split configuration.

4.9 LightGBM Results

The LightGBM classifier achieved 95.19% accuracy using the polynomial kernel.

4.10 Linear SVM Results

The Linear SVM classifier achieved a peak accuracy of 98.08% using the polynomial kernel.

4.11 MLP Classifier Results

The MLP classifier achieved a maximum accuracy of 95.19%.

4.12 Naive Bayes Results

The Naive Bayes classifier performed significantly better with the linear kernel compared to the polynomial kernel.

4.13 Quadratic Discriminant Analysis Results

The QDA classifier achieved a maximum accuracy of 98.08% using the polynomial kernel.

4.14 Random Forest Results

The Random Forest classifier achieved 98.08% accuracy using the linear kernel under the 80-20 split.

4.15 XGBoost Results

The XGBoost classifier achieved 99.04% accuracy using the polynomial kernel.

4.16 AdaBoost Results

The AdaBoost classifier achieved 98.08% accuracy using the linear kernel.

4.17 Comparative Analysis

Among all 15 algorithms tested, KSVM with a linear kernel achieved the best overall performance with 100% accuracy.

The results clearly show that machine learning algorithms combined with Kernel PCA can effectively improve diabetes detection accuracy.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This research successfully developed a machine learning-based diabetes prediction system using Kernel PCA and multiple classification algorithms.

The study tested 15 machine learning algorithms using a dataset containing 521 patient records and 17 health-related features. After preprocessing and dimensionality reduction, the models were evaluated using multiple train-test split configurations.

Among all models, Kernel Support Vector Machine (KSVM) with a linear kernel achieved the highest accuracy of 100% under the 80-20 train-test split. Other models such as XGBoost and CatBoost also demonstrated excellent performance with 99.04% accuracy.

The results confirm that machine learning techniques can significantly assist healthcare professionals in early diabetes diagnosis and decision-making.

5.2 Future Work

Future improvements may include:

- Testing the model on larger real-world hospital datasets.
- Integrating the system with electronic health records.
- Developing a web or mobile application for real-time prediction.
- Applying deep learning techniques for improved analysis.
- Implementing explainable AI methods for better interpretability.

Advantages of the Proposed System

- High prediction accuracy
- Early diabetes detection
- Reduced dependency on laboratory testing
- Faster diagnosis process
- Useful for rural healthcare environments
- Supports healthcare professionals in decision-making

Limitations of the Study

- Limited dataset size
- Results may vary with different datasets
- Requires preprocessing for accurate predictions
- Real-time hospital implementation was not performed

Applications

- The proposed system can be applied in:
- Hospitals and healthcare centers
- Rural healthcare clinics
- Medical research organizations
- Healthcare mobile applications
- Smart healthcare systems